Lab 7: Fuzzing for Input Validation Bugs

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Application: library_fines.py

Test file: test_library_fines.py

Repository: L6_L7_14_library_fines (Lab 7)

1. Running the buggy application

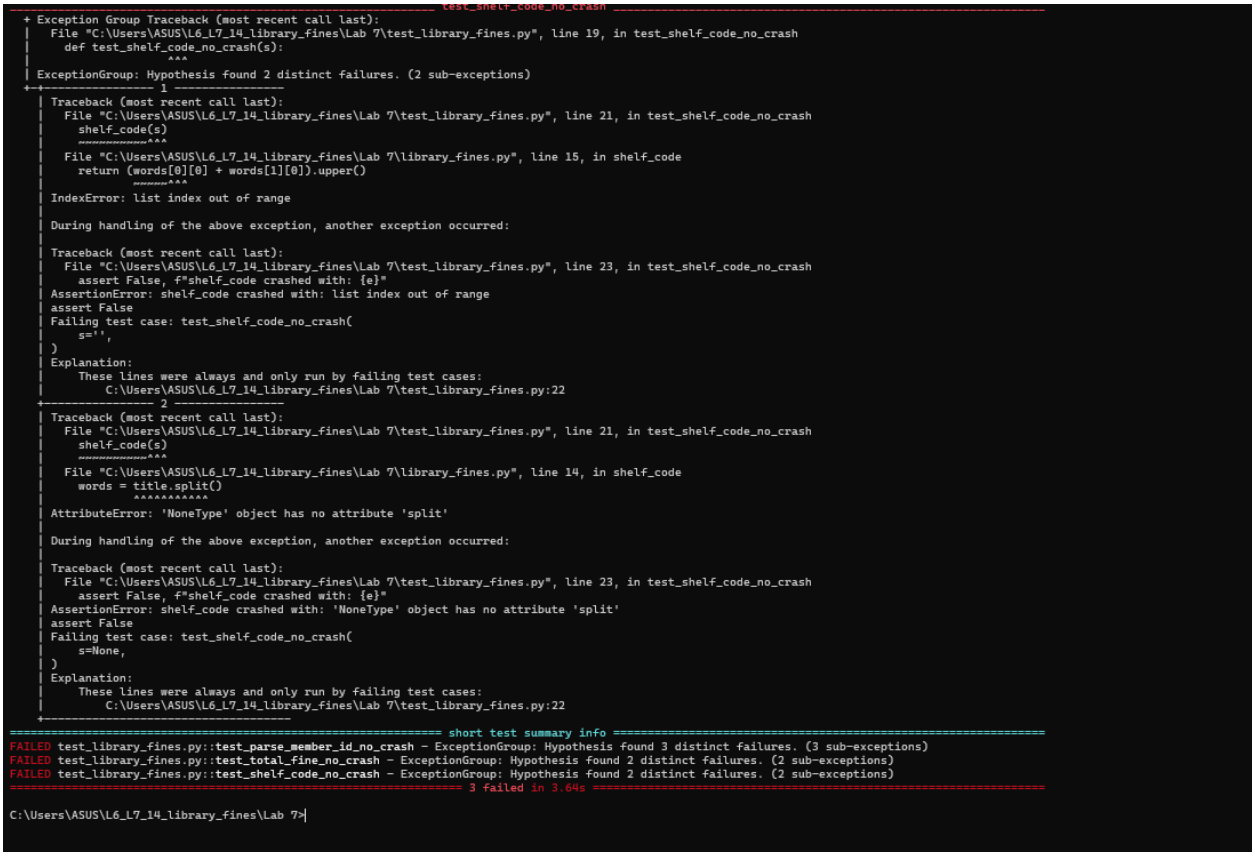
The original library_fines.py crashes with an unhandled ValueError when the member ID does not contain a hyphen.

```
C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU1023
Traceback (most recent call last):
  File "C:\Users\ASUS\L6_L7_14_library_fines\Lab 7\library_fines.py", line 34, in <module>
    main()
    ~~~~^^
  File "C:\Users\ASUS\L6_L7_14_library_fines\Lab 7\library_fines.py", line 20, in main
    result = parse_member_id(user_input)
  File "C:\Users\ASUS\L6_L7_14_library_fines\Lab 7\library_fines.py", line 2, in parse_member_id
    member_type, number = member_id.split('-')
    ^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>
```

2. Running the fuzz tests against the buggy code

All three Hypothesis tests fail. Hypothesis reduces each failure to the smallest input that reproduces it.



3. Running the fixed application

Valid input still produces the correct result.

```
C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU-1520
Member Type and Number: ('STU', 1520)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,20,30
Total Fine: 44

==== Running shelf_code ====
Enter a book title with at least two words: The Great Gatsby
Shelf Code: TG

C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>
```

The input that previously crashed now produces a validation message instead of an exception.

```
C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>python library_fines.py
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): STU1023
Invalid member ID. Expected the form TYPE-NUMBER, e.g. STU-1023.

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): 10,20,30
Total Fine: 44

==== Running shelf_code ====
Enter a book title with at least two words: The Great Gatsby
Shelf Code: TG

C:\Users\ASUS\L6_L7_14_library_fines\Lab 7>
```

4. Re-running the fuzz tests against the fixed code

All three tests pass.

